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Comparison of Clinical Outcomes for Patients with Large Choroidal Melanoma after Primary Treatment with Enucleation or Proton Beam Radiotherapy

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Key Words
Choroidal melanoma · Melanoma · Ocular tumours · Radiation oncology · Radiotherapy

Abstract
Purpose: To evaluate survival and clinical outcome for patients with a large uveal melanoma treated by either enucleation or proton beam radiotherapy (PBRT). ***Procedures:*** This retrospective non-randomized study evaluated 132 consecutive patients with T3 and T4 choroidal melanoma classified according to TNM stage grouping. ***Results:*** Cumulative all-cause mortality, melanoma-related mortality and metastasis-free survival were not statistically different between the two groups (log-rank test, p = 0.56, p = 0.99 and p = 0.25, respectively). Eye retention of the tumours treated with PBRT at 5 years was 74% (SD 6.2%). In these patients at diagnosis, 73% of eyes had a best-corrected visual acuity (BCVA) of 0.1 or better. After 12 and 60 months, BCVA of 0.1 or better was observed in 47.5 and 32%, respectively. ***Conclusion and Message:*** Although enucleation is the most common primary treatment for large uveal melanomas, PBRT is an eye-preserving option that may be considered for some patients.

Introduction

Uveal melanoma is the most common primary malignancy of the eye, although it is still a rare cancer with an estimated incidence in Italy of 2–4 cases per million people [1]. Enucleation has long been considered the reference treatment for uveal melanoma.

However, in 1978, Zimmerman et al. [2] cast doubt on the benefits of enucleation and suggested that enucleation may actually promote tumour metastasis.

Thus, over the last 30 years there has been an increased interest in alternative forms of therapy for this condition and the development of a number of conservative therapies. In this context the Collaborative Ocular Melanoma Study validated the brachytherapy as a valuable alternative to enucleation in the treatment of medium-sized uveal melanoma [3]. Proton beam radiotherapy (PBRT) has been used for the treatment of uveal melanoma since 1975 [4]. The use of PBRT has some advantages with respect to brachytherapy. Indeed PBRT allows for a homogeneous dose distribution within the whole tumour volume with a sharp decrease in dose outside the target.

Larger tumours are associated with an increased risk of local failure with conservative therapies [5–7], and for

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this reason enucleation is the treatment of choice in larger tumours. Nevertheless, certain factors may favour eye-conservative treatments. In our experience, such factors include useful vision in the affected eye, poor vision in the contralateral eye or a strong desire for ocular retention.

In this retrospective study, we analysed the survival rates of patients affected with large uveal tumours treated by enucleation or PBRT. The main outcome measures were melanoma-related mortality, all-cause mortality, metastasis-free survival, frequency of eye retention and final visual acuity.

Methods

Patients and Inclusion Criteria

We performed a retrospective review of the clinical records of patients with choroidal melanoma treated by enucleation or PBRT at the Ocular Oncology Service in Genoa between August 1996 and July 2007. In this study, we enrolled patients with unilateral choroidal tumours classified as T3 and T4 tumours, according to the seventh TNM classification [8].

We excluded previously treated tumours, diffuse, ring or multifocal tumours, or tumours judged to be predominantly ciliary body melanoma. In addition, we excluded patients with metastatic disease at the time of treatment, with other primary tumours or a history of cancer.

Clinical Evaluation

Tumour dimensions were determined by a combination of clinical assessments and standardized A- and B-scan ultrasonography (CineScan S, Quantel Medical Inc.). All patients were referred for a complete metastatic melanoma workup prior to and after any therapy (i.e. liver function tests and liver ultrasonography every 6 months and chest radiography every 12 months).

The same surgeon (C.M.) performed enucleation or scleral clip insertion for tumour localization in all patients. PBRT was performed at the Centre Antoine-Lacassagne Cyclotron Biomedical, in Nice, France. A total dose of 60-Gy equivalents was delivered in 4 fractions over 4 consecutive days for all patients. PBRT-treated patients were followed by clinical examination, indirect ophthalmoscopy, retinography and ocular ultrasound (standardized A and B scans) 1 month after treatment and every 6 months thereafter. Local tumour recurrence was defined as any documented tumour growth (base or thickness) after PRBT detected by A-scan or B-scan ultrasonography or serial comparison of fundus photographs. Eyes of PBRT-treated patients who suffered either local recurrence of the tumour or complications were subsequently enucleated or retreated with PBRT.

We performed a comparative analysis on melanoma-related mortality, all-cause mortality as well as metastasis rate. Methods of determining survival status included follow-up examination or contact with patients and patients' families. Outcomes were then coded as follows: alive without known melanoma metastases, alive with uveal melanoma metastases, death due to uveal melanoma metastases or to causes other than melanoma. If the date of metastasis onset was unknown, we used the date of melanoma-related death in the analysis of metastasis-free survival.

Table 1. Characteristics of 132 patients with a large uveal melanoma who underwent enucleation or PBRT

	Enucleation group (n = 62)	PBRT group (n = 70)	p value
Male sex	34 (61%)	43 (55%)	0.95
Mean age ± SD, years	66.7 ± 14.5	62.7 ± 14.1	0.10
Right eyes	30 (48%)	35 (50%)	0.98
Tumour thickness, mm			<0.01
Mean ± SD	12.0 ± 2.8	9.8 ± 1.6	
Range	6.8–21	6.2–13.5	
Largest basal diameter, mm			0.23
Mean ± SD	14.4 ± 4.5	15.2 ± 2.7	
Range	5–25	8.3–21.3	

The χ^2 test was used for proportions and Student's test for means.

Statistical Analysis

Tests of independence on the characteristics of the PBRT and enucleation groups were performed by means of univariate and multivariate two-sample t tests. Survival analysis on death from all causes, death from melanoma-related causes and metastasis-free survival was performed by Kaplan-Meier analysis [9]. Univariate prognosis factors were estimated by Kaplan-Meier analysis using the log-rank test. The Cox proportional hazards model analysis [10] was then applied, including age, sex, tumour thickness, maximum basal tumour diameter and ciliary body invasion as prognostic factors. The level of statistical significance was set at 0.05 (two-sided t test). All analyses were performed using the *survival* and *stats* packages of the open source statistical package R (R project – www.r-project.org).

Results

A total of 132 patients met the inclusion criteria and were included in the study. The enucleation group comprised 62 patients treated by enucleation between December 1999 and November 2007. The PBRT group comprised 70 patients treated between August 1996 and June 2007. Table 1 shows the clinical characteristics of the two groups.

The mean follow-up time was similar between groups (p = 0.08): 45.5 months (SD 21.6) in the enucleation group and 53.4 months (SD 29.3) in the PBRT group. The median follow-up was 42.4 months (CI 8.2–82.5) in the enucleation group and 52.9 (CI 14.6–107.6) months in the PBRT group (p = 0.18).

Figure 1 shows the Kaplan-Meier survival curves of patients treated with enucleation or PBRT for all-cause

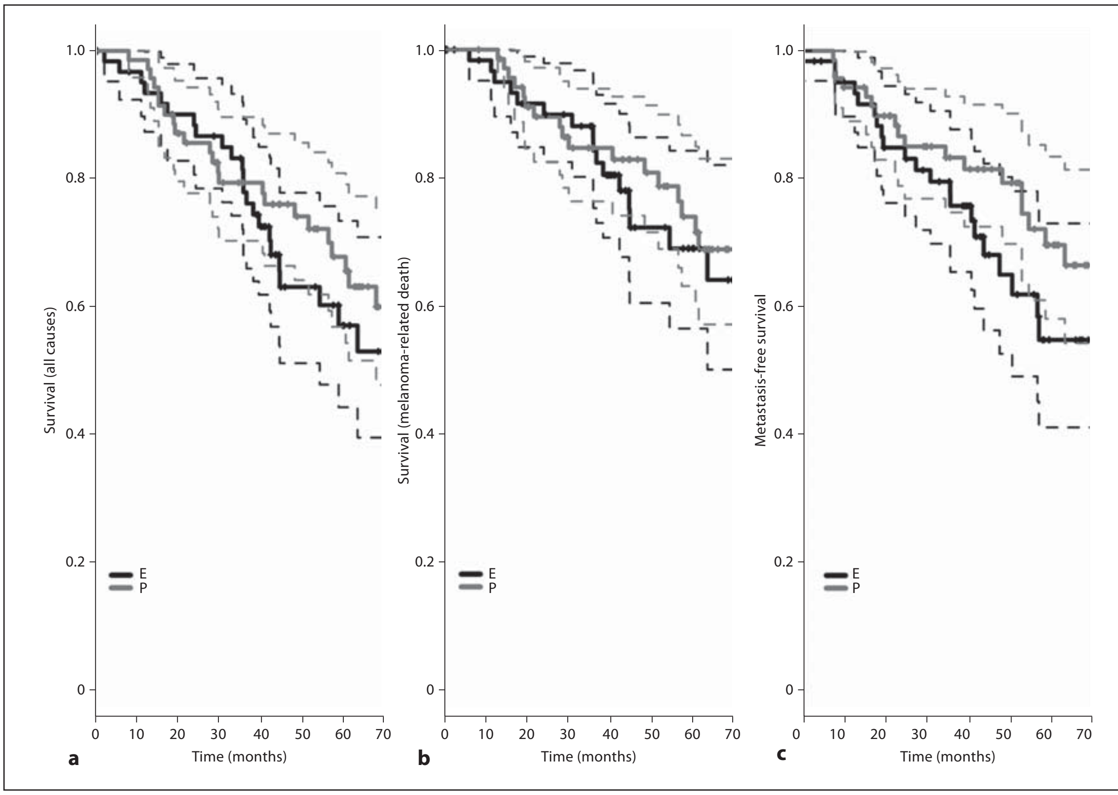

Fig. 1. a Kaplan-Meier survival curves comparing all-cause mortality of patients treated with enucleation or PBRT. **b** Kaplan-Meier survival curves comparing melanoma-related mortality of patients treated with enucleation or PBRT. **c** Kaplan-Meier survival curves for metastasis-free survival of patients treated with enucleation or PBRT. E = Enucleation; P = PBRT.

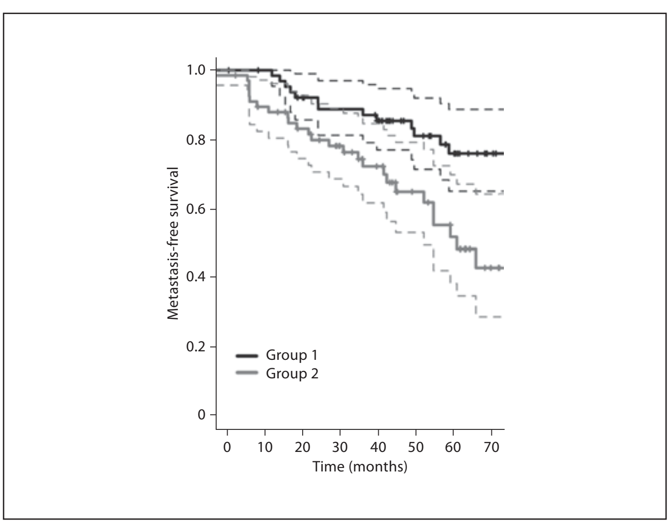

Fig. 2. Kaplan-Meier survival curves for metastasis-free survival of patients treated with enucleation (E) and PBRT (P) differing for maximum base diameter. Group 1 includes tumors with a diameter ≤14.7 mm. Group 2 includes tumors with a diameter >14.7 mm.

mortality, melanoma-related mortality and metastasis-free survival. The Kaplan-Meier survival curves between the two groups were not significantly different ($p = 0.56$, $p = 0.99$ and $p = 0.25$, respectively).

In enucleated patients, the 5-year all-cause mortality was 43% (SD 7%), the 5-year melanoma-related mortality was 39% (SD 7%) and the 5-year metastasis-free survival was 55% (SD 8%). In PBRT patients, the 5-year all-cause mortality was 34% (SD 6%), the 5-year melanoma-related mortality was 38% (SD 6%) and the 5-year metastasis-free survival was 72% (SD 6%). A Cox proportional model revealed no effect associated with the type of treatment (enucleation or PBRT) on metastasis-free survival after correcting for age ($p = 0.075$), tumour thickness ($p = 0.84$) and sex ($p = 0.32$). Conversely, the maximum base diameter factor seems to have an effect, having a p value of 0.044. Therefore, we proceeded with a Kaplan-Meier analysis grouping the data according to the diameter (higher or lower than the median diameter: 14.7 mm). The resulting curve is pictured in figure 2 ($p = 0.005$).

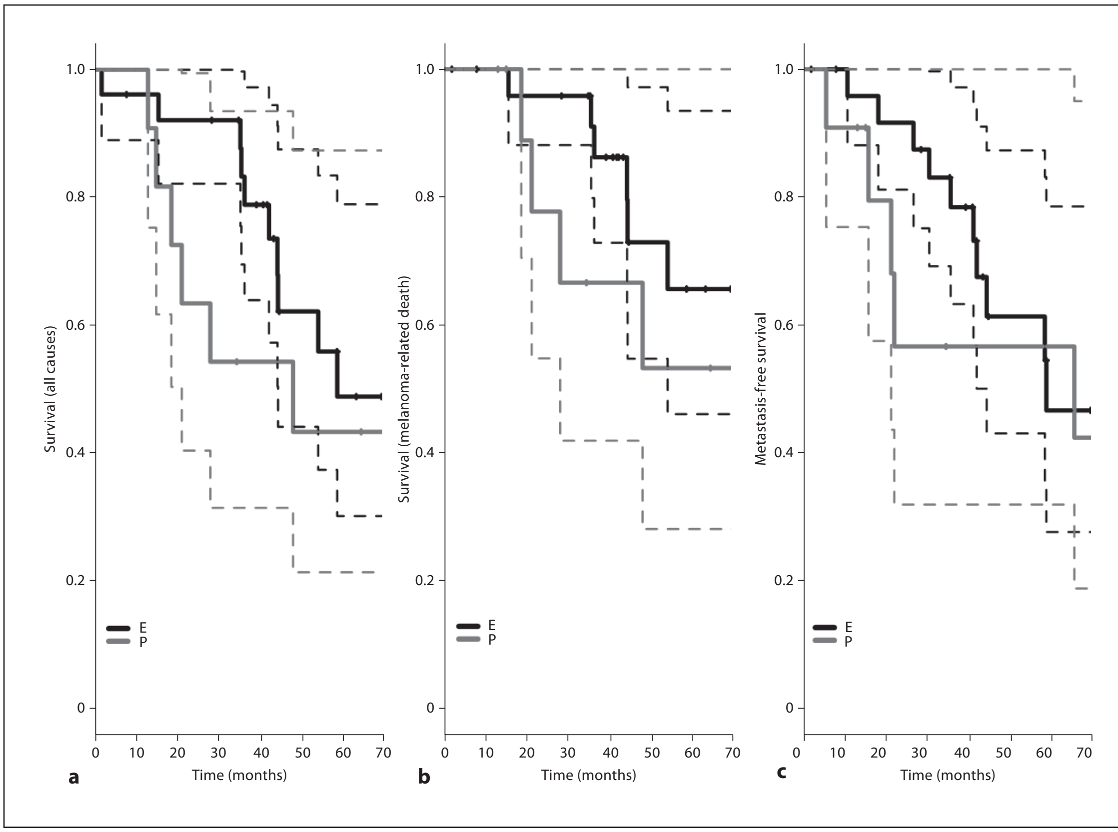

Fig. 3. a Kaplan-Meier survival curves comparing all-cause mortality of T3 patients treated with enucleation or PBRT. **b** Kaplan-Meier survival curves comparing melanoma-related mortality of T3 patients treated with enucleation or PBRT. **c** Kaplan-Meier survival curves for metastasis-free survival of T3 patients treated with enucleation or PBRT. E = Enucleation; P = PBRT.

Following the first part of the statistical analysis, we considered separately the T3 and T4 samples (table 2 shows the TNM classification of the two groups) and performed the same statistical protocol, discriminating between treatments.

In T3 patients we obtained a mean follow-up time of 43.9 months (SD 21.8) in the enucleation group and 55.1 months (SD 29.2) in the PBRT group ($p = 0.04$). The median follow-up was 41.5 months (CI 9.7–78.2) in the enucleation group and 53.6 months (CI 15.0–112.0) in the PBRT group ($p = 0.09$). Similar to what we did in the analysis of the entire data set, we evaluated the Kaplan-Meier survival curves of patients treated with enucleation or with PBRT for all-cause mortality, melanoma-related mortality and metastasis-free survival (fig. 3). The Kaplan-Meier survival curves between the two groups were not significantly different ($p = 0.31$, $p = 0.72$ and $p = 0.27$, respectively).

Table 2. TNM staging of choroidal melanomas

Stage	Enucleation group (n = 62)	PBRT group (n = 70)
T3	36	59
T4	26	11

In T4 patients we obtained a mean follow-up time of 47.6 months (SD 21.6) in the enucleation group and 44.6 months (SD 29.3) in the PBRT group ($p = 0.77$). The median follow-up was 43.9 months (CI 10.0–81.5) in the enucleation group and 34.6 months (CI 14.2–87.4) in the PBRT group ($p = 0.5$). Figure 4 shows the Kaplan-Meier survival curves of patients treated with enucleation or with PBRT for all-cause mortality, melanoma-related mortality and metastasis-free survival. The Kaplan-Meier survival

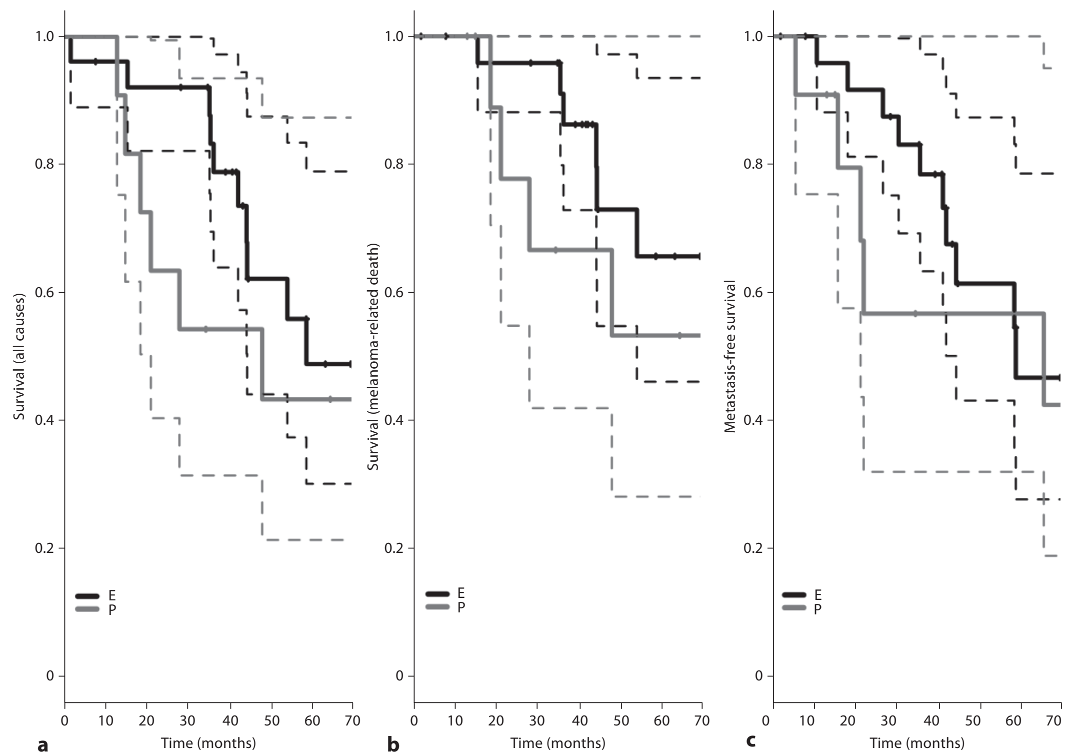

Fig. 4. a Kaplan-Meier survival curves comparing all-cause mortality of T4 patients treated with enucleation or PBRT. **b** Kaplan-Meier survival curves comparing melanoma-related mortality of T4 patients treated with enucleation or PBRT. **c** Kaplan-Meier survival curves for metastasis-free survival of T4 patients treated with enucleation or PBRT. E = Enucleation; P = PBRT.

curves between the two groups were not significantly different ($p = 0.21$, $p = 0.32$ and $p = 0.51$, respectively).

Overall, 10 out of the 70 (14%) tumours treated with PBRT recurred locally with a mean time of 27.7 months (SD 29.4). One was controlled with a second course of PBRT and 9 were managed with secondary enucleation.

Eye retention was 74% (SD 6.2%) over 5 years. The cumulative eye retention rate after PBRT is shown in figure 5. Eye retention was better in the T3 than in the T4 group ($p < 0.01$).

Visual acuity after PBRT is summarized in figure 6 by a histogram plot showing best-corrected visual acuity (BCVA) before treatment and 12, 24, 36 and 48 months after. At diagnosis 73% (51/70) of eyes had a BCVA of 0.1 or better. After 12, 24, 36, 48 and 60 months, BCVA of 0.1 or better was observed in 47.5% (29/61), 39% (19/49), 31% (12/39), 34% (10/29) and 32% (7/22) of patients, respectively.

Discussion

This non-randomized study indicated that the survival of the PBRT group after treatment was not worse than that in the enucleation group in terms of all-cause mortality, melanoma-related mortality and metastasis-free survival. These results were similar not only when all patients were included, but also for the T3 and T4 subgroups.

Over 5 years, about 75% of patients treated with PBRT saved the eye and, of them, about one third preserved useful vision.

We are confident of the presented results also because the sample size of our study is relatively large (n = 132). Moreover, our data indicate largest tumour diameter as the main predictor of metastatic disease, reaffirming the results of previous studies [11, 12].

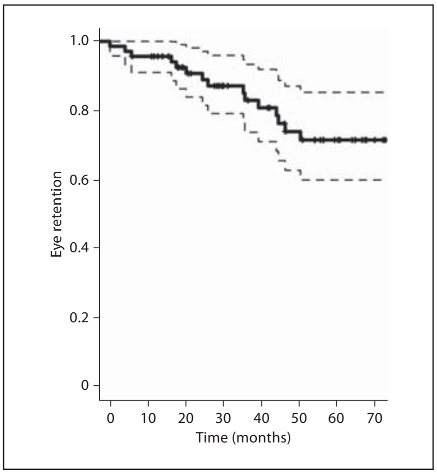

Fig. 5. Cumulative eye retention rate after PBRT.

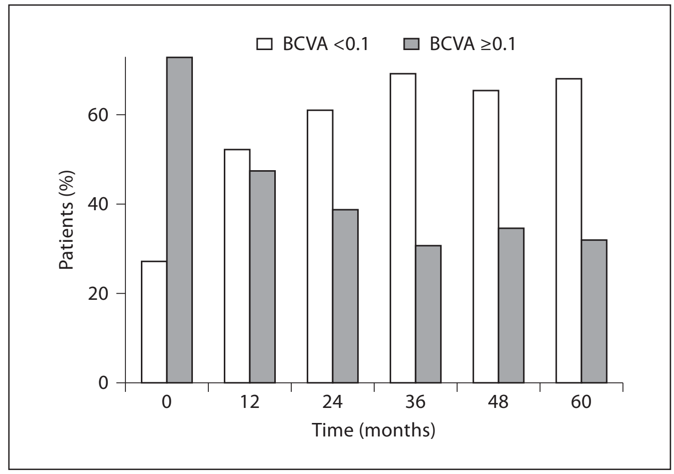

Fig. 6. Histogram of BCVA in patients treated with PBRT before therapy and 12, 24, 36, 48 and 60 months after treatment.

Our study could be integrated with information about chromosomal abnormalities. We aim to present a more complete study in the future incorporating such genomic information which was recently introduced in our centre as a routine procedure; moreover a standard practice in patients with intra-ocular neoplasms has yet to be defined [13].

For what concerns the enucleation group, we found a 5-year survival rate similar to that described in the Collaborative Ocular Melanoma Study large choroidal melanoma clinical trial [14].

Recent reports investigated the use of brachytherapy and tumour resection as an eye-sparing treatment for large melanoma. The plaque size is limited to the amount of radiation that can be tolerated by the sclera, as well as by the physical constraints of the ocular and orbital anatomy. In particular, it is difficult to place the plaque for tumours located close to the optic nerve, and these cases have a higher risk of developing radiation maculopathy or optic neuropathy, as well as an increased risk of marginal failure. In a larger series of cases, the majority of tumours treated by plaque radiotherapy had a maximum diameter of less than 18 mm [15–17].

Another option could be tumour resection; this procedure requires hypotensive anaesthesia, a very experienced surgery team and the possibility of additional vitreoretinal surgery. Tumour resection may be less feasible in patients who are poor surgical candidates [18, 19].

PBRT is considered a possible alternative to enucleation for large tumours that are unmanageable with brachytherapy. Because of the homogeneous dose distribution achievable with proton beams, this technique was also considered to be an alternative to brachytherapy for large tumours located close to the optic disc or the macula, or both.

Gragoudas et al. [20] demonstrated that patients with large tumours treated by PBRT had more than twice the risk of recurrence as patients with smaller tumours. In the present study, tumour relapse was observed at a higher frequency in the PBRT group than reported in our previous study (14 vs. 8.4%) [21], an event that we correlate with the larger dimensions of tumours (mean thickness of 9.8 mm and mean maximum diameter of 15.2 mm vs. mean thickness of 6.2 mm and mean maximum diameter of 14.2 mm).

In a previous study, Conway et al. [22] evaluated ocular retention and visual retention following PBRT in 21 extralarge uveal melanomas. Compared with the present study, choroidal melanomas had a smaller mean tumour thickness (8.6 mm), a larger mean tumour basal diameter (18.7 mm) and ciliary body melanomas were also included. At 24 months, probabilities of local control and eye control were 67 and 54%, respectively; BCVA of 0.1 or better was preserved at 24 months in 25% of patients.

Although our study confirms a poor long-term prognosis of maintaining a useful visual acuity for irradiated eyes with a large uveal melanoma, many patients prefer short-term conservation of vision to immediate blindness from primary enucleation; we believe that PBRT can be a useful option in these patients who are unwilling to undergo radical treatment.

References

1 Virgili G, Gatta G, Ciccolallo L, Capocaccia R, Biggeri A, Crocetti E, Lutz JM, Paci E, EUROCARE Working Group: Incidence of uveal melanoma in Europe. Ophthalmology 2007;114:2309–2315.
2 Zimmerman LE, McLean IW, Foster WD: Does enucleation of the eye containing a malignant melanoma prevent or accelerate the dissemination of tumour cells? Br J Ophthalmol 1978;62:420–425.
3 Collaborative Ocular Melanoma Study Group: The COMS randomized trial of iodine-125 brachytherapy for choroidal melanoma. V. Twelve-year mortality rates and prognostic factors. COMS Report No. 28. Arch Ophthalmol 2006;124:1684–1693.
4 Gragoudas ES, Goitein M, Verhey L, Munzenreider J, Suit HD, Koehler A: Proton beam irradiation: an alternative to enucleation for intraocular melanomas. Ophthalmology 1980;87:571–581.
5 Augsburger JJ, Corrêa ZM, Trichopoulos N: An alternative hypothesis for observed mortality rates due to metastasis after treatment of choroidal melanomas of different sizes. Trans Am Ophthalmol Soc 2007;105:54–59.
6 Collaborative Ocular Melanoma Study Group: Development of metastatic disease after enrollment in the COMS trials for treatment of choroidal melanoma. Collaborative Ocular Melanoma Study Group Report No. 26. Arch Ophthalmol 2005;123:1639–1643.
7 Egger E, Zografos L, Schalenbourg A, Beati D, Böhringer T, Chamot L, Goitein G: Eye retention after proton beam radiotherapy for uveal melanoma. Int J Radiat Oncol Biol Phys 2003;55:867–880.
8 Edge SD, Byrd DR, Carducci MA, Compton CC, Fritz AG, Greene FL: AJCC Cancer Staging Manual, ed 7. New York, Springer, 2009.
9 Kaplan E, Meier P: Non-parametric estimation from incomplete observation. J Am Stat Assoc 1958;53:457–481.
10 Cox DR: Regression models and life-tables. J R Stat Soc B 1972;34:187–220.
11 Shammas HF, Blodi FC: Prognostic factors in choroidal and ciliary body melanomas. Arch Ophthalmol 1977;95:63–69.
12 Damato B, Coupland SE: A reappraisal of the significance of largest basal diameter of posterior uveal melanoma. Eye (Lond) 2009;23:2152–2160.
13 Eide N, Walaas L: Fine-needle aspiration biopsy and other biopsies in suspected intraocular malignant disease. Acta Ophthalmol 2009;87:588–601.
14 Collaborative Ocular Melanoma Study Group: The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma. IV. Ten-year mortality findings and prognostic factors. COMS Report No. 24. Am J Ophthalmol 2004;138:936–951.
15 Shields CL, Naseripour M, Cater J, Shields JA, Demirci H, Youseff A, Freire J: Plaque radiotherapy for large posterior uveal melanomas (≥8 mm thick) in 354 consecutive patients. Ophthalmology 2002;109:1838–1849.
16 Puusaari I, Heikkonen J, Summanen P, Tarkkanen A, Kivelä T: Iodine brachytherapy as an alternative to enucleation for large uveal melanomas. Ophthalmology 2003;110:2223–2234.
17 Kaiserman N, Kaiserman I, Hendler K, Frenkel S, Pe'er J: Ruthenium-106 plaque brachytherapy for thick posterior uveal melanomas. Br J Ophthalmol 2009;93:1167–1171.
18 Damato B, Foulds WS: Indications for trans-scleral local resection of uveal melanoma. Br J Ophthalmol 1996;80:1029–1030.
19 Puusaari I, Damato B, Kivelä T: Transscleral local resection versus iodine brachytherapy for uveal melanomas that are large because of tumour height. Graefes Arch Clin Exp Ophthalmol 2007;245:522–533.
20 Gragoudas E, Li W, Goitein M, Lane AM, Munzenrider JE, Egan KM: Evidence-based estimates of outcome in patients irradiated for intraocular melanoma. Arch Ophthalmol 2002;120:1665–1671.
21 Mosci C, Mosci S, Barla A, Squarcia S, Chauvel P, Iborra N: Proton beam radiotherapy of uveal melanoma: Italian patients treated in Nice, France. Eur J Ophthalmol 2009;19:654–660.
22 Conway RM, Poothullil AM, Daftari IK, Weinberg V, Chung JE, O'Brien JM: Estimates of ocular and visual retention following treatment of extra-large uveal melanomas by proton beam radiotherapy. Arch Ophthalmol 2006;124:838–843.